Evgenii Zheltonozhskii

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Randl

Skills and expertise

Deep learning	PyTorch (3 years), TensorFlow (1 year)
Deep learning research	Published 11 papers and preprints in multiple fields of DNN, including self-supervised learning and adversarial defences, since 2017. Wide knowledge of current trends in computer vision.
Software development	Python (4 years), C++ (5 years, including 2 years of C++17), Linux, LaTeX, git, jupyter, pandas, numpy

Experience

August – October 2020 **Research Intern**, *Snap Research*, Los Angeles (remote), Creative vision group.

2016 – 2020 **Research Assistant**, *Technion*, Haifa, Professor Alex Bronstein's group.

- Research in Deep Neural Networks and their implementation in hardware;
- Investigation of compression methods and their impact on DNN performance;
- Implementation and reproduction of recent DL algorithms and papers.

June – August 2017 **Google Summer of Code Participant**, *OpenCV*.
GPU enabled deep learning framework: introducing GPU support for tiny-dnn, C++14 header-only deep learning library

2013 – 2016 **Freelancer**, *Upwork*.
C++, Python programmer

Open source contribution

2019 – 2020 **TensorFlow**.
Implementation of differentiable eigendecomposition in TensorFlow

2016 – 2018 **tiny-dnn**.
Maintainer of tiny-dnn, header only, dependency-free deep learning framework in C++14

Education

Spring 2020 - present **MSc in Computer Science**, *Technion – Israel Institute of Technology*, Haifa.

- Advisor: Professor Alex Bronstein
- Research interests: Self-supervised deep learning, geometrical DL, complex computer vision problems, applications of DL in physics, Bayesian DL, bias in ML systems;
- Teaching experience: "Intro to Machine Learning", "Deep Learning on Computational Accelerators", organization of seminar in Deep Learning

2016 – 2020 **BSc in Computer Science and BSc in Physics and Mathematics**, *Technion – Israel Institute of Technology*, Haifa, GPA *91.70*.

- Participant of Excellence Program;
- President's List (top 3% of students): Spring 2018;
- Dean's List (top 15% of students): Winter 2016, Spring 2017, Winter 2018, Spring 2019, Winter 2019;
- Research projects in condensed matter physics:
  - "Renormalization Group Modelling of Interface between $p + ip$ Superconductors";
  - "Experimental Realization of Anomalous Floquet-Anderson Insulators";
- ICPC semifinals: SWERC 2018 – honorable mention, SWERC 2019 – bronze medal (11th place).

Summer 2018 **DeepBayes**, *Summer school on Bayesian methods in deep learning*.

Publications

[1] **Evgenii Zheltonozhskii**, Chaim Baskin, Alex M. Bronstein, and Avi Mendelson. "Self-Supervised Learning for Large-Scale Unsupervised Image Clustering". In: *arXiv pre-print* (Aug. 2020).

[2] Alex Karbachevsky, Chaim Baskin, **Evgenii Zheltonozhskii**, Yevgeny Yermolin, Freddy Gabbay, Alex M. Bronstein, and Avi Mendelson. "HCM: Hardware-Aware Complexity Metric for Neural Network Architectures". In: *arXiv pre-print* (Apr. 2020).

[3] **Evgenii Zheltonozhskii**, Chaim Baskin, Yaniv Nemcovsky, Brian Chmiel, Avi Mendelson, and Alex M. Bronstein. "Colored Noise Injection for Training Adversarially Robust Neural Networks". In: *arXiv pre-print* (Mar. 2020).

[4] Yaniv Nemcovsky, **Evgenii Zheltonozhskii**, Chaim Baskin, Brian Chmiel, Alex M. Bronstein, and Avi Mendelson. "Smoothed Inference for Adversarially-Trained Models". In: *arXiv pre-print* (Nov. 2019).

[5] Yury Nahshan, Brian Chmiel, Chaim Baskin, **Evgenii Zheltonozhskii**, Ron Banner, Alex M. Bronstein, and Avi Mendelson. "Loss Aware Post-training Quantization". In: *arXiv pre-print* (Nov. 2019).

[6] Chaim Baskin, Brian Chmiel, **Evgenii Zheltonozhskii**, Ron Banner, Alex M. Bronstein, and Avi Mendelson. "CAT: Compression-Aware Training for bandwidth reduction". In: *arXiv pre-print* (Sept. 2019).

[7] Brian Chmiel, Chaim Baskin, Ron Banner, **Evgenii Zheltonozhskii**, Yevgeny Yermolin, Alex Karbachevsky, Alex M. Bronstein, and Avi Mendelson. "Feature Map Transform Coding for Energy-Efficient CNN Inference". In: *2020 International Joint Conference on Neural Networks (IJCNN)*. July 2020, pp. 1–9. DOI: 10.1109/IJCNN48605.2020.9206968. URL: https://arxiv.org/abs/1905.10830.

[8] Yochai Zur, Chaim Baskin, **Evgenii Zheltonozhskii**, Brian Chmiel, Itay Evron, Alexander M. Bronstein, and Avi Mendelson. "Towards Learning of Filter-Level Heterogeneous Compression of Convolutional Neural Networks". In: *ICML 2019 AutoML Workshop* (Apr. 2019). URL: http://arxiv.org/abs/1904.09872.

[9] Chaim Baskin, Natan Liss, Yoav Chai, **Evgenii Zheltonozhskii**, Eli Schwartz, Raja Giryes, Avi Mendelson, and Alexander M. Bronstein. "NICE: Noise Injection and Clamping Estimation for Neural Network Quantization". In: *arXiv pre-print* (Oct. 2018).

[10] Chaim Baskin, Eli Schwartz, **Evgenii Zheltonozhskii**, Natan Liss, Raja Giryes, Alexander M. Bronstein, and Avi Mendelson. "UNIQ: Uniform Noise Injection for the Quantization of Neural Networks". In: *arXiv pre-print* (Apr. 2018).

[11] Chaim Baskin, Natan Liss, **Evgenii Zheltonozhskii**, Alexander M. Bronstein, and Avi Mendelson. "Streaming Architecture for Large-Scale Quantized Neural Networks on an FPGA-Based Dataflow Platform". In: *2018 IEEE International Parallel and Distributed Processing Symposium Workshops, IPDPS Workshops 2018, Vancouver, BC, Canada, May 21-25, 2018*. May 2018, pp. 162–169. DOI: 10.1109/IPDPSW.2018.00032. URL: https://arxiv.org/abs/1708.00052.